



Unresolved Issues in Supply Chain Network Design

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Abstract. Designing a supply chain network provides the basic structure for supply chain operations, where the network is a major element in a firm's competitiveness and a significant area of capital investment. With the boundary-spanning scope encouraged by supply chain management, the question is whether existing analyses used for network design are adequate. The general answer is yes, because the number of echelons that the analysis must span is self-limiting to a relative few. However, there remain unresolved issues in network design analysis and in the models used to support the analysis that can enhance its accuracy, appropriateness, and acceptability to practitioners. Issues discussed relate to the development of better models, the treatment of the data needed by them, and the need for comparing the various models for network design.

Key Words. facility location, logistics network configuration, supply chain management, network design

1. Introduction

Facility location, network configuration, supply chain design, and logistics system strategy are terms that have been used to describe what has been known to researchers as the warehouse location problem. Among the many problems encountered in strategic logistics planning, this is perhaps the most significant. Over the years, top management has sponsored probably more studies dealing with this problem than any other of a logistical nature. The need for this type of study has been driven by change resulting from increasing customer expectations about logistics service, lengthening distribution lines as global markets are penetrated, and shifting demand patterns brought about by merger and acquisition programs. Since reconfiguring the facilities of a supply channel frequently can result in a logistics cost reduction of 5 to 15% while maintaining or improving customer service, the benefits have been

significant enough to deserve the attention of high-level executives. Consider some of the reasons why companies undertake a network redesign.

- A major acquisition by a division of a large corporation resulted in two network designs, one for the current division and one for the acquired company. Re-evaluation was needed to meld the network configurations into a single, efficient design.
- An automobile manufacturer was faced with importing vehicles to the U.S. market from Latin America and Europe. Latin America production was new, which changed the flow of product that originally was only from Europe and brought new products into the sales mix. Revised product sourcing required determining ports of entry and delivery staging areas that would optimize the product flow patterns to dealers from plants of limited capacity.
- A rail car leasing company repairs the undercarriages of the cars in its fleet. Cars are brought back from rail yards to repair locations on a scheduled or emergency repair basis. The number and location of the repair hubs had never been rationalized, thus an incentive was provided for their evaluation.

Facility location provides the infrastructure for the supply chain and establishes the base from which operating economies are realized. It is a problem of balancing conflicting costs (production/purchase, warehousing, inventory, and transportation) with customer service goals and the associated revenue contribution. Historically, logistics system design has been limited to that portion of the logistics channel under direct control of the company. Concepts surrounding supply chain management encourage making cost tradeoffs across company boundaries. This is extending the scope of the analysis and placing increased demands on mathematical tools used to solve such problems.

In this article, the impact of the supply chain movement on facility location analysis will be examined to see the challenges posed to traditional location methodology. In addition, other issues that affect the quality of location analysis will be discussed. The intent is to suggest directions for location analysis and model development that will enhance the accuracy and scope of network design.

2. The Network Design Problem

In the boldest sense, the network design problem is one of configuring the nodal points on a product flow network that range from the sources of raw materials to the points of final consumption. It is to determine the number, size, and location of terminals, ports, plants, warehouses, pool points, cross docks, or any point in the transportation network where product is stopped momentarily on its way to the end consumer. A particular network might look like that shown in Fig. 1. The objective of network design is to maximize a financial ratio such as ROLA.

$$\text{ROLA} = \frac{\text{Revenue} - \text{Costs}}{\text{Investment}}$$

where ROLA = Return on logistics assets, Revenue = Contribution that network design makes to sales, Costs = Operating costs of the supply channel, Investment = Capital required to establish the facilities in the network configuration.

Location has been a popular area of research, with methods described in the literature as early as the mid 1800s by land economists such as von Thünen (1842) and Weber (1909), and more recently by regional economists such as Palander (1935), Lösch (1940), Hoover (1957), Greenhut (1956), Isard et al. (1960), and Isard (1968). Many of the methods they describe were of a conceptual nature and have been referred to as “chart-compass-ruler techniques” because of their limited scope to deal with problems of a practical size. However, they are useful as aids to understanding the general nature of location problems. Mathematicians have continued with the search for location methods that can find optimal answers to location questions while expanding the scope and complexity of the problems that can be treated. Brandeau and Chiu (1989) and Drezner (1995) have summarized the methods that are available. More recently, Ballou and Masters (1999) surveyed the commercial models used by firms to analyze their facility location problems. The most popular methodology found in commercial use is some form of

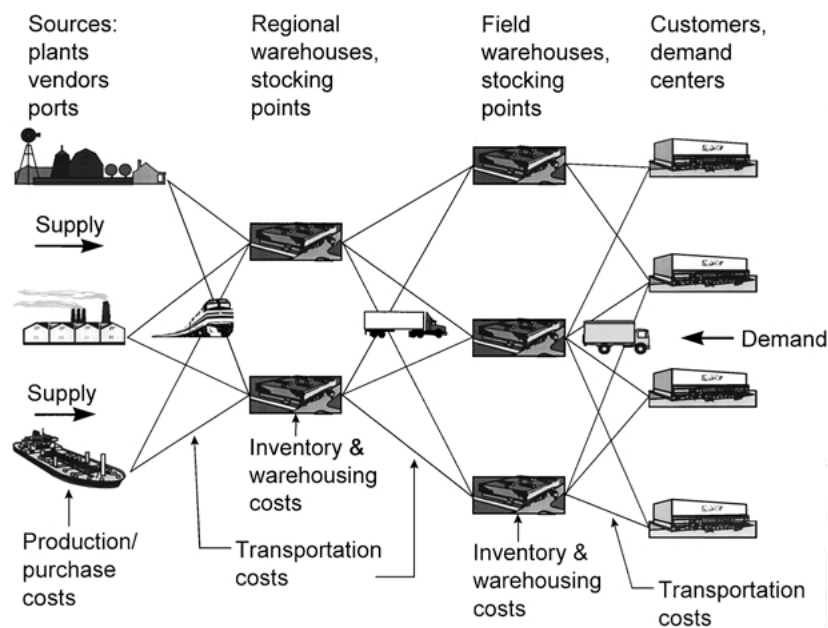


Fig. 1. A typical supply chain network with associated costs.

mathematical programming, such as linear programming or mixed integer programming. Heuristics are still employed to aid these methods in the search for optimal or nearly optimal locations.

Although substantial gains have been made in the size of location problems that can be treated mathematically, mainly due to increased computer power, there remain numerous issues that need to be addressed. Some deal with the scope of the problem being solved, but many have to do with the data issues surrounding the location problem. It is the intent of this paper to highlight these needed directions in the hope that further research will be encouraged.

3. The Supply Chain and Facility Location

Supply chain is a recent term used to describe the entire product flow process, from the source of raw materials to end consumption, regardless of who owns the various stages of the process. Although not new to logisticians, supply chain management is promoting the idea of planning across the legal boundaries of firms within the supply channel. This potentially requires developing multi-firm network configurations that mesh well with each other. A supplier's warehouses may be located in concert with a buyer's plant, which they supply. Independent retailer outlets may be co-located with their supplying distributor/vendor location. Some firms may locate a plant or warehouse near a large JIT customer to provide the necessary service, but in general there appears to be little evidence that firms are coordinating their location decisions with other channel members. Typically, upstream sources and downstream demand are taken as given in the location decision. This remains the case even though most commercial-grade location models have the capability of handling the many echelons required in an extensive supply chain (Ballou and Masters, 1999).

There are several possible reasons for the general lack of location coordination among channel members. First, information sharing is required with its attendant problems, such as confidentiality. Second, the sales importance of a downstream channel member may be small relative to the total customer base such that there is little incentive to co-locate. Third, there can be a problem of product identification for upstream suppliers with the end products of downstream buyers. Including additional channel members in the location

decision may not be needed except in those limited situations where economic and service incentives are obvious. A certain level of decoupling of the supply channel may be reasonable in the case of location decisions.

4. Developmental Issues in Facility Location

Facility location remains a significant problem in supply chain management. Large-scale facility location models have been developed that assist in designing complex networks. Yet, there are a number of unresolved issues that keep facility analysis and their associated models from being as effective as they might be. The mathematics of modeling seem reasonably well developed, and computing power is great enough to computationally handle most practical location problems. Those issues that can increase the accuracy, applicability, and scope of current facility location analysis are discussed.

4.1. Aggregate, hierarchical, and channel planning

Warehouse location is generally viewed as a strategic planning problem where the facility location is found within a broad geographic area. However, available models used for aggregate planning don't locate the specific facility site. Whereas aggregate location planning can be effectively based on economic factors, site selection involves many subjective factors such as local laws, community attitudes, availability of utilities, quality of schools, and proximity to transportation links. To the extent that these subjective factors can be quantified, including them in the aggregate analysis would be a step forward.

The location problem has a hierarchical nature. In addition to its strategic orientation, operational considerations are sometimes brought into the analysis. For example, the scheduling of facility labor, the routing of privately-owned trucks, or the setting of inventory levels (given the throughput assigned to a facility by the location analysis) might be considerations within network design. These are difficult issues to include in network modeling, mainly because they are of a temporal nature. Warehouse location models are fundamentally spatially based and do not handle temporal issues particularly well. Temporal matters have been treated through preprocessing of data for spatial models or through the iterative use of multiple models such

as mathematical programming and simulation. Effective models that handle spatial and temporal issues well in a combined way remain elusive.

Taking a broad view of the supply chain, facility location may involve the location of source facilities such as plants, vendors, and ports; intermediate facilities such as warehouses, consolidation points, terminals, and cross dock facilities; and retail outlets. The methodologies on which location decisions are based throughout the channel have been quite different. Source point and intermediate point location decisions have generally followed a cost analysis with customer service, and therefore contribution to sales, being taken as a constraint on the analysis. On the other hand, retail location is driven by the potential for generating sales, costs being a secondary consideration (Ghosh and McLafferty, 1987). Combining these disparate approaches into a seamless analysis, or single model, so that a channel-wide network design can be accomplished would be ideal.

4.2. Standards for data representation

Describing models and analyses that are comprehensive enough to treat an entire supply chain overlooks the unresolved issues that remain within the existing models. One of those issues surrounds the way in which data are represented in a location analysis. Even though developers of models may claim mathematical optimality, lacking standards for data representation leads to poor comparability among methods and variability in location results that depend very much upon the judgment of an individual analyst. Consider a few of the problems that can occur.

Facility costs. Costs for company-owned facilities are accumulated as accounting records. Individual accounts may show the costs for such categories as labor, trash removal, electricity, natural gas, and parking lot paving. Since most analyses require that such costs be grouped as annual fixed, variable handling, and sometimes storage, matching the accounts to these categories can vary widely among analysts. The result can be that different well-meaning analysts can arrive at strikingly different network designs.

Facility capacity. Capacity is, and will remain, a definitional problem. Limiting facility throughput is a basis for arguing that mathematical programming is the appropriate method for treating location problems (Geoffrion and Van Roy, 1979). Mathematical programming can handle capacity constraints in an exact way. In practice, capacity is rarely a fixed number

against which to optimize. Adjusting production shifts, product mix, overtime, and number of workers all can change the throughput from a fixed facility size. As with facility costs, a precise definition of capacity is needed.

Product families. A location problem frequently involves thousands of product items grouped into relatively few families. Broad guidelines suggest that products having the same general distribution characteristics, i.e., they flow through the supply channel in the same manner, can effectively be treated in the same way. There are no uniformly applied standards for product classification.

Customer service. Customer service in facility location is particularly difficult to handle mainly because the analysis is geographically based, i.e., customers and facilities are identified by geocodes. This leads to treating customer service as some form of distance between sourcing points and customers. With speed, shipping time may be represented. However, representing customer total order cycle time that includes inventory availability remains illusive for this type of analysis. Some location models prune network links with unacceptable service times through data preprocessing, but don't include time within their optimizing algorithm. Representing customer service as total order cycle time, i.e., the time from when a customer requests a product to when it is received, is preferable to using delivery distance as a surrogate for delivery time.

4.3. Accuracy associated with data aggregation

It is common practice in location analysis to aggregate a number of the data elements for the purpose of simplifying the analysis, decreasing computational time, and reducing the amount of data handling required in a location analysis. Research has provided little information as to the error that may be caused by data aggregation. A few of the critical areas where further research is needed are discussed.

Product families. Creating product groups, as previously mentioned, is the first element in location analysis where data are aggregated. A company can have thousands of items in its product line. Modeling each is prohibitive and generally unnecessary since many flow through a supply channel in the same manner. However, product grouping requires that the family be represented in terms of average statistics, such as value, production costs, warehouse handling costs, and demand. In a word, the cost characteristics of an individual product are lost in the averages of multi-product

groupings without knowing the solution effects of doing so.

Demand clusters. A company will sell to thousands of geographically dispersed customers or purchase from numerous vendors. It is common practice to cluster these customers/vendors into relatively few geographical zones, such as 3-digit zip codes (Bender, 1985). The risk of doing this is transportation costs are calculated to the cluster center rather than to/from the actual customer/vendor address, causing a potential cost estimating error. Some research on the error has been conducted for truck shipments (Ballou, 1994), but no research reports have been found for other transport modes. Since transportation costs are one of the important driving forces in facility location, the error might significantly affect the location outcomes.

Transport rate-estimating curves. In a network analysis, transportation rates are needed between all points within all echelons. When shipments over a period of time are made by different transport modes and within various weight breaks, there can be millions of point-to-point transport rates needed. From a data collection and computational standpoint, it is attractive to construct transport rate estimating curves where rates are estimated from the distance between two points, since a computer can calculate the distance and solve the rate equation faster than looking for the rate in a large rate table. Fitting a linear least squares curve to rates plotted against distances gives a high degree of correlation, often greater than 0.90, for the published rates in the truck tariffs in most cases (Ballou, 1991), as illustrated in Fig. 2. Published rates in other countries also show this same high degree of

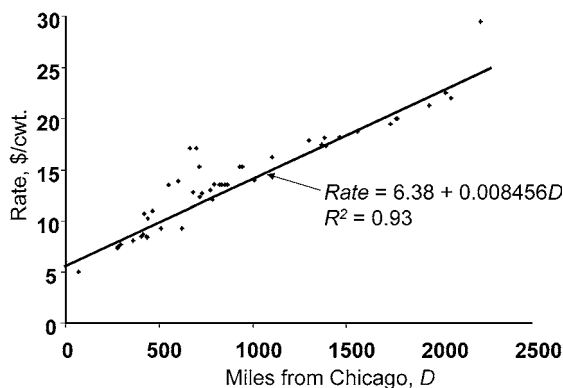


Fig. 2. Truck rate estimating curve for selected destinations from Chicago.

rationalization. Although using the rate curves in location analysis appears to be reasonable, the costing error and its impact on location decisions have not been well established.

Travel distances. Actual traveled distances are needed in location analysis to approximate transportation rates and determine distance-based customer service. Common practice is to approximate actual distances from location coordinates, typically latitude and longitude, and apply a multiplier (called a circuitry factor) to the straight-line distances so as to approximate actual travel distances. A circuitry factor is the average ratio of actual traveled distance to straight-line distance. Although the calculation of straight-line, or great circle, distances can be quite accurate, the circuitry factors throughout the world can show great variability (Ballou, Rahardja, and Sakai, 2000). The preferred, or frequently traveled, route through a road, rail, or other network is subject to interpretation, which can cause inconsistencies among data sources. Perhaps improved methods can be found for distance estimation, although the current approach is simple and computationally efficient.

Inventory levels. Estimating inventory levels throughout a supply chain network as a result of demand allocation is a requirement of good location analysis. Some general relationships for making the estimation have been developed from inventory theory and actual data (Ballou, 1981, 2000), but documentation of the error in practice remains. Combining items into an aggregate inventory-throughput relationship can lose the effect of individual inventory policies of individual items and may be a significant source of estimating error.

4.4. Extensions to scope

Facility location analysis, especially involving modeling, is an incomplete work. Some of the limitations are obvious from the previous discussion, but several additional ones are presented here. They deal with extending the scope of the analysis and enriching location models.

Profit contribution. Some years ago, Shycon (1982) argued that companies are in the business to generate profit, growth, and a reasonable return on investment, yet much of supply chain system design is focused on cost reduction. His suggestions have largely gone unheeded, even though today many location models include price and claim to have the capability of profit maximization. The ability to generate a reasonable

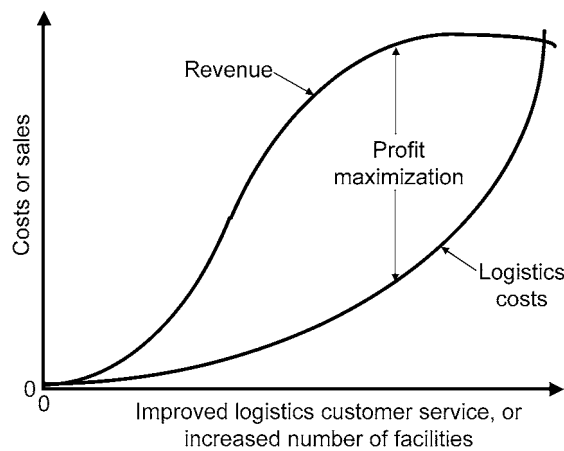


Fig. 3. Logistics revenue contribution as a factor in location.

sales-service (revenue) relationship as shown in Fig. 3 that will explain quantitatively the extent to which facility locations impact sales is a key, but missing, ingredient. Without it, location analysis fundamentally minimizes costs while treating customer service as a constraint. The revenue effects of location are largely ignored, except by Ho and Perl (1995). With it, location becomes part of the firm's strategy to maximize ROLA. However, extending the scope of location analysis to include revenue effects depends on developing methods for accurately determining sales-logistics customer service relationships, not just including such relationships within a location model.

Inventories in location analysis. There appears to be no standard way to handle the "inventory consolidation effect" in location analysis. The inventory consolidation effect refers to the changes in inventory levels that occur when the number of facilities in a supply chain network, and the demand allocations to them, changes. Since inventory-carrying cost is in tradeoff with transportation cost, inventory-carrying cost should be an integral part of the search for the optimal number of locations. However, too many analyses set facility inventory levels *after* locations are determined rather than *during* the search process. This is probably due to most models used for warehouse location being of the mixed integer linear programming type (Ballou and Masters, 1999) that require the relationship between facility inventory levels to be linear, when the relationship may be nonlinear as shown in Fig. 4 (recall the square root law of inventory consolidation). Some analysts argue that they can treat nonlinear inventory costs in a piecewise

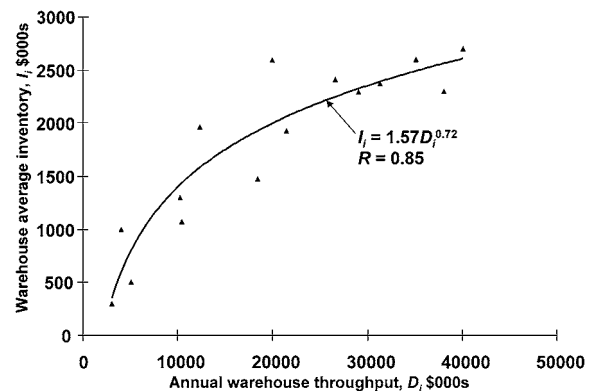


Fig. 4. Example of inventory levels increasing nonlinearly with warehouse throughput, as developed from turnover ratios of a company's warehouses.

linear fashion, but there is little evidence of doing this routinely, probably because of the resulting significant increase in problem size.

Transportation. In location analysis, it is generally assumed that product movement is directly from an origin point to a destination point. This is a reasonable assumption when simple common carrier moves are made, since rates are quoted between point pairs. However, when transportation involves the peddle routes of private carriers or for-hire carriers offer transit privileges where the rates are a result of combining the volumes for more than one stop, the problem of transport costing becomes much more difficult. The assigning of customer demand to facilities is based on transportation rates that can only be determined *after* the customer assignments are known. The location and transportation problems can be solved separately and combined in an iterative fashion. Some research has been conducted to combine these two effects (Bookbinder and Reese, 1988; Perl and Daskin, 1984), but much work remains to bring the preliminary results to a practical level.

Selected product sourcing. It has previously been pointed out that data aggregation is an important element of location analysis to control problem size. However, when creating product families from thousands of product line items, the uniqueness of the sourcing location for intermediate facilities can be lost. That is, when a facility such as a warehouse is sourced from plants, it is assumed that any particular plant can produce all products within the family. If some products can only be sourced from particular plants, then

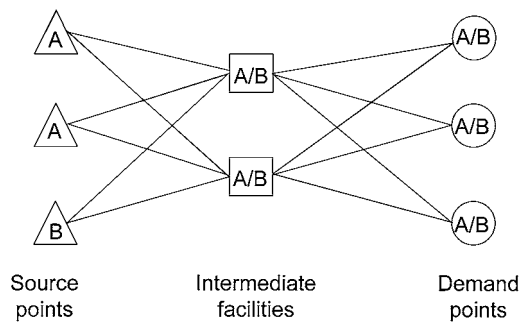


Fig. 5. Selected sourcing among products within a product family.

misallocation can occur during optimization, as shown Fig. 5. The recourse is to disaggregate the product family until all the remaining products in the family can be sourced from an available plant. Of course, a way is sought to formulate the location problem so that selected sourcing can be achieved within the heterogeneous product family.

Demand uncertainty. Uncertainty of customer demand in a location problem is rarely a consideration in model design. Static models dominate and demand uncertainty is treated in location analysis through sensitivity analysis. Since network design is usually handled as a strategic planning problem, demand uncertainty is not quite as critical to analysis as it is in many operating level problems. However, some research would be useful to determine practical ways of treating uncertain demand and under what circumstances it is appropriate to use the methods.

Flexible costing. Allocation models suffer from the problem that their solutions are based on costs known and fixed during the solution-searching process. Since costs are typically derived from an assumed allocation, the costs used to generate a solution may change as a result of the solution—the “chicken-and-egg” dilemma. This is frequently a problem in location analysis where the demand assignments to facilities are based on production, facility, and transportation costs that are influenced by economies of scale. Revising the demand assignments can change these costs, making the assignments to facilities no longer optimal. A procedure needs to be developed where a model’s costs can be adjusted while the demand assignments are being developed so that the costs are appropriate for the allocations.

Spatial vs. temporal issues. Some years ago, it was observed that spatial location analysis did not adequately account for decisions that were strongly based

on time (Heskett, 1965, 1966). Inventory stocking, transportation mode selection, shipment size, and order cycle time are issues only crudely handled in a spatial analysis, where spatial analysis dominates location methodology. On the other hand, supply channel simulations treat these temporal issues nicely and provide information about facility inventory levels and transportation rates based on a mix of shipments inbound and outbound from a facility. However, this information that is important to location analysis depends on location analysis results to provide the demand assignments to the facilities in the network. Again, this is the “chicken-and-egg” dilemma.

Simulators provide unique information for network design, but they do not make good locators due to their impracticality to search for good locations. An attempt was made to combine the temporal and spatial aspects of network design (Bowersox et al., 1972), but special purpose supply channel simulators have not been commercially popular. Even the idea of using location analysis and simulators in an iterative manner has not been fashionable. The ability is needed to generate information about these temporal issues, so perhaps they can be included effectively within the spatial analysis. If research cannot produce spatial models with this broader scope, then improved ways for combining locators and simulators should be explored.

Static vs. dynamic analysis. Solving a location problem for a particular period of time (static analysis) does not capture the true nature of network design. If the time period is the most recent year for which data are available, the solution is probably already out of date. New facilities, or even modified existing facilities, cannot be brought online immediately. Of course, analysis can be directed to the demand and costs for some future year so as to address the timing issue. However, location over time is a dynamic problem where the optimal network design may be different in each year. Since there are fixed costs associated with changing from one configuration to the next, merely switching to the current-year optimal design is not necessarily optimal over the entire planning horizon. A dynamic analysis is needed where the optimal set of moves from one network design to another is charted. Several researchers have addressed the problem (Ballou, 1968; Sweeney and Tatham, 1976; Wesolowsky and Truscott, 1975; Van Roy and Erlenkotter, 1982; Bowersox, 1972), but little of this work has found practical application, probably because running times for optimal procedures, e.g., dynamic programming, are long for even small problems.

The importance of dynamic location suggests that it should be part of the scope of location analysis and that practical methods for its treatment are needed.

Continuous vs. discrete models. The popular method of analysis is to use discrete models, typically in the form of mathematical programming models. Locations are selected from a specified list of candidate sites. On the other hand, continuous location models, variously known as center of gravity, centroid, or first moment methods, determine the coordinate locations anywhere within the demand region. This has appeal to practitioners who want the analysis to determine the locations rather than to be limited to a potentially biased set of candidate sites, as is the case with mathematical programming. However, continuous location methods have long been dismissed as a general approach to practical problems, although there are limited cases where the method is appropriate. The reasons seem to be the lack of robustness (lack of multiple echelons, capacity constraints on facilities, customer service restrictions, and a variety of cost configurations) that can be achieved in model design and still realize an optimal solution in a reasonable time. Having said this, such models still have a place in location analysis.

Continuous location models can be used as site suggestors. They can identify locations for examination by the more robust discrete models. Location analysis would be enhanced if such models could be incorporated into the computational processes of the discrete models.

5. Method Comparison

Perhaps one of the most important unresolved issues in supply chain network design is the comparison of one location analytical approach to another. Over the years, analysts, especially model developers, have claimed superior performance for their approach. Without a doubt, recipients of the analytical results would like to know which approach is "best." Too often "best" is argued on the basis of mathematical optimality for the solved problem when such issues as (1) accuracy of problem representation, (2) time to collect and prepare data for analysis, (3) solution time, (4) presentation features, (5) transparency of the methodology and results, (6) flexibility, and (7) database/model interface can be of equal or greater importance. There is no known comparison of commercial-grade location

models, although a few years ago Andersen Consulting attempted but did not complete such a task. Complexity and fairness top the list of factors why further comparison efforts have not been tried. Model developers may continue to argue superiority on the basis of mathematical sophistication when end users of an analytical approach equally value other, subjective factors as well (Ballou and Masters, 1993, 1999). The differences need to be resolved.

6. Conclusions

Contemporary supply chain management encourages planning across the legal boundaries of supply channel members. This has the potential of leading to reduced costs and/or improved customer service as compared to planning strictly within those logistics activities under the immediate control of the firm. Configuring the logistics network seems an obvious candidate for this boundary-spanning planning. However, this system-wide planning will likely be limited to relatively few echelons because (1) information sharing among channel members will bring its attendant problems, (2) the importance of the firm to its upstream and downstream partners may not be great enough to encourage cooperative location planning, and (3) the identification of the product over multiple echelons becomes blurred. Although location models and their associated analyses can handle many echelons of a network, the practical scope of such analyses will be limited to relatively few members of the channel, probably no more than one upstream and one downstream member.

Although a typical location analysis supported by a location model is probably adequate for most supply channel configuration problems encountered, there remain important developmental issues that can increase the accuracy, applicability, and scope of location analysis. First, the scope of the analysis should be extended to deal with the hierarchical nature of the location problem. Second, standards need to be set on how costs, capacities, product groupings, and customer service are defined and prepared for analysis. Third, the error associated with the degree of data aggregation associated with product grouping, demand clustering, transport rate estimating, distance estimating, and inventory level estimating should be determined for the purpose of benefiting from reduced problem size and complexity while retaining reasonable accuracy in the problem description. Fourth, current commercial models for

location analysis would benefit from extending their scope by including profit contribution, inventory consolidation effects, multi-stop transportation, selected product sourcing, demand uncertainty, flexible costing, temporal effects, dynamic location effects, and continuous modeling types.

Another important unresolved issue is which model solution approach is best? Users selecting a model want to know. Model developers want to know. Defining reasonable ground rules and inviting location model developers to participate in a fair test with impartial judging would provide valuable information to guide location model use and development.

In many respects, current location analysis serves us well and is likely to be adequate in the emerging environment of broad channel planning promoted in supply chain management. However, there remain many research issues surrounding data representation, problem scope, and model testing that can significantly improve the practical application of location analysis and the models used to support it.

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